

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	13
Experiment Name	Conduct a Z-Test Using Python

DESCRIPTION

To perform the experiment: Conduct a Z-Test Using Python.

CODE

```
import numpy as np
from scipy.stats import norm

sample = np.array([152,148,151,149,147,153,150,148,152,149,
                  151,150,149,152,151,148,150,152,149,150,
                  148,153,151,150,149,152,148,151,150,153])

mu0 = 150
sigma = 2.0 # assumed known population standard deviation
alpha = 0.05
n = len(sample)
z = (sample.mean() - mu0) / (sigma / np.sqrt(n))
p_value = 2 * (1 - norm.cdf(abs(z)))
print("Sample mean:", round(sample.mean(), 2))
print("Z statistic:", round(z, 4))
print("P-value:", round(p_value, 4))
print("Reject H0" if p_value < alpha else "Fail to reject H0")
```

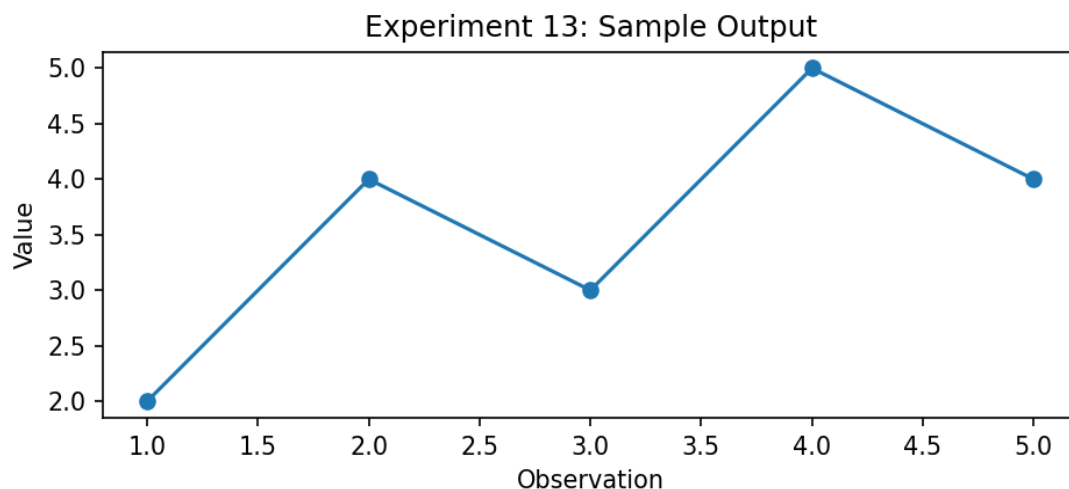
SAMPLE INPUT

Sample input: 30 measurements; hypothesized population mean 150; known population standard deviation assumed to be 2; alpha = 0.05.

OUTPUT — SAMPLE CONSOLE OUTPUT

Sample mean, Z statistic, p-value and decision printed.\nDecision is based on alpha = 0.05.

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature